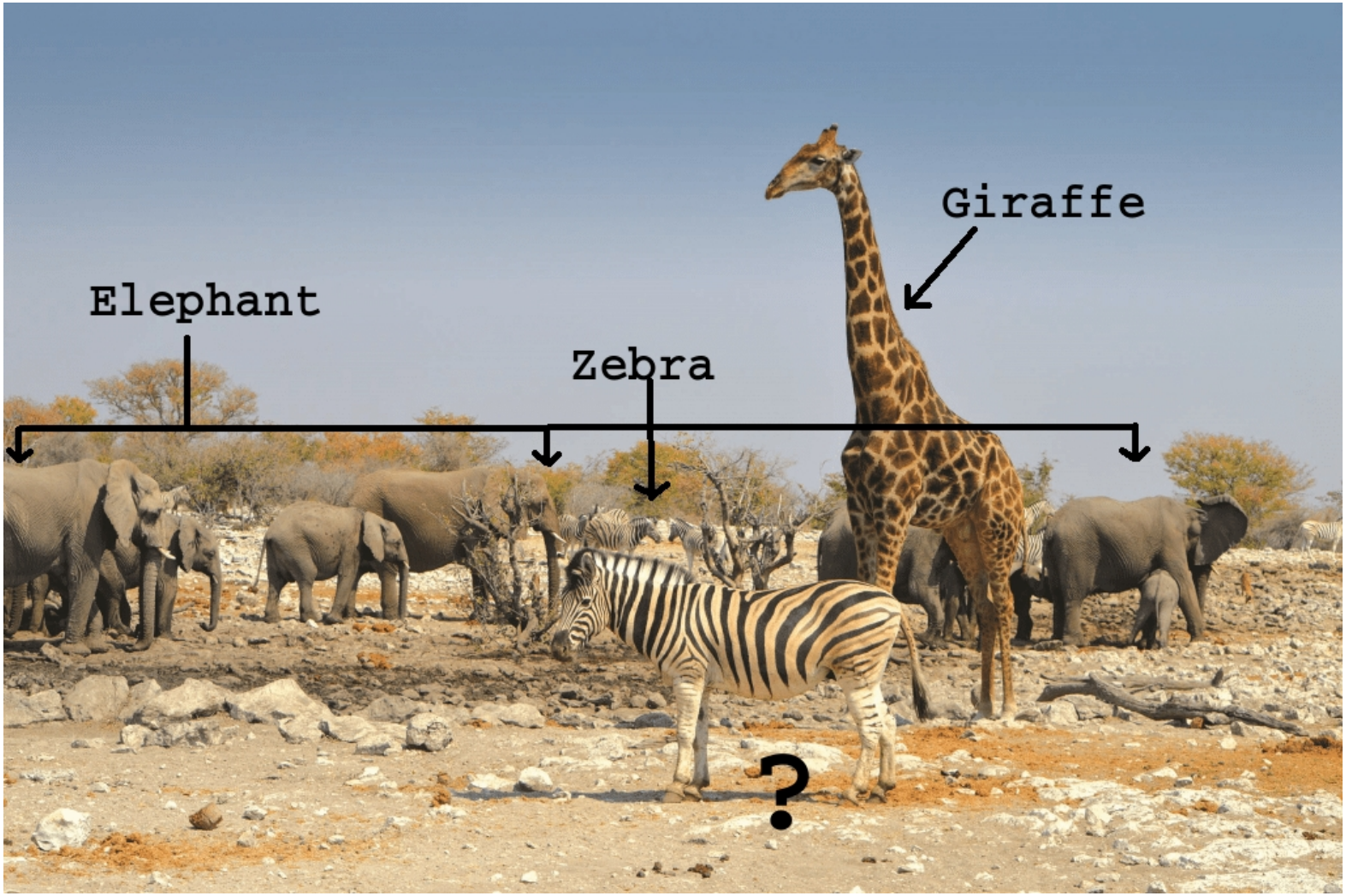
CLASSIFICATION

Classification Example



Classification

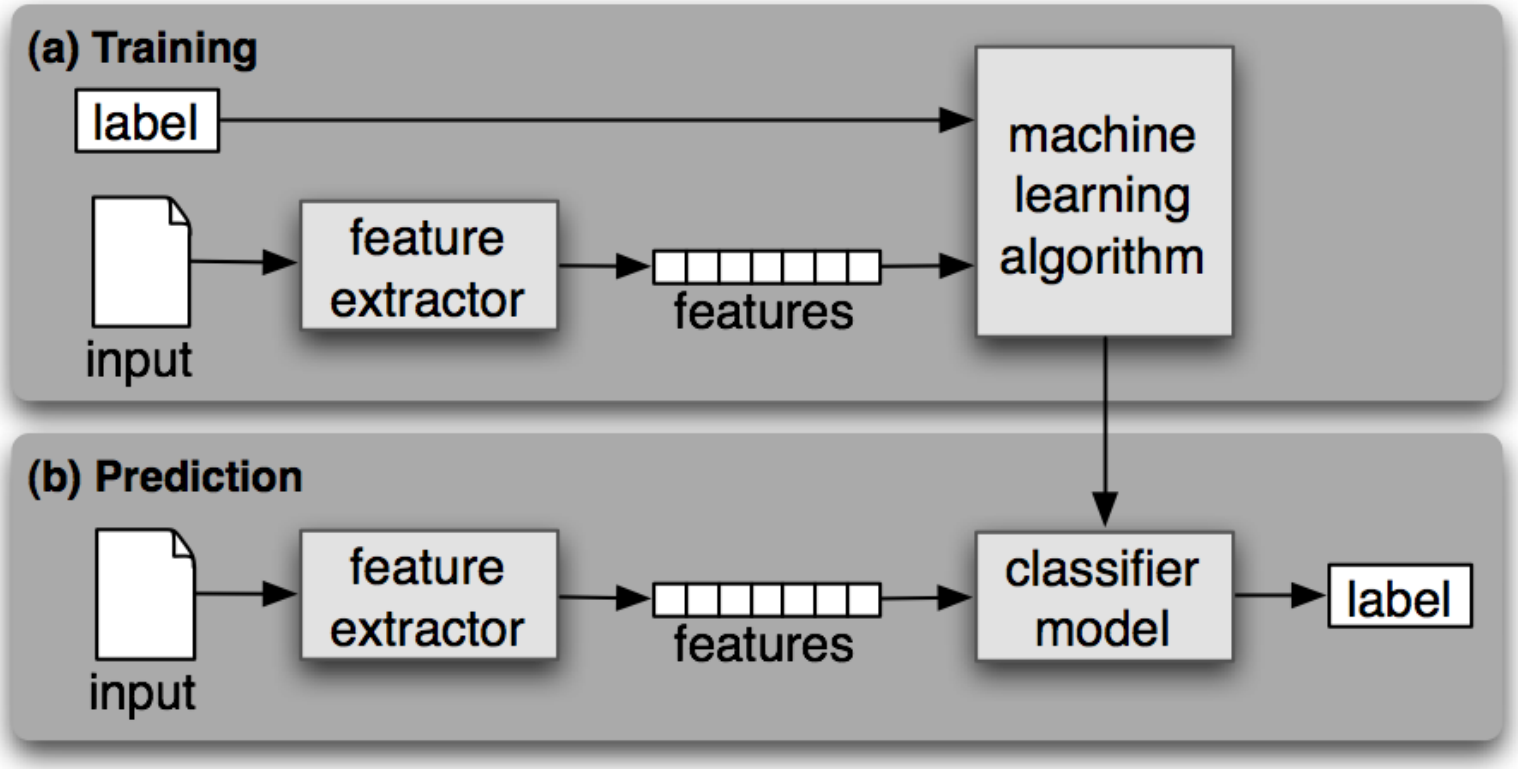
Definition

An applications in which the *training data* comprises examples of the input vectors (known as supervised learning problems) and the aim is to assign each input vector to one of a finite number of discrete categories (classes) is called a classification problem

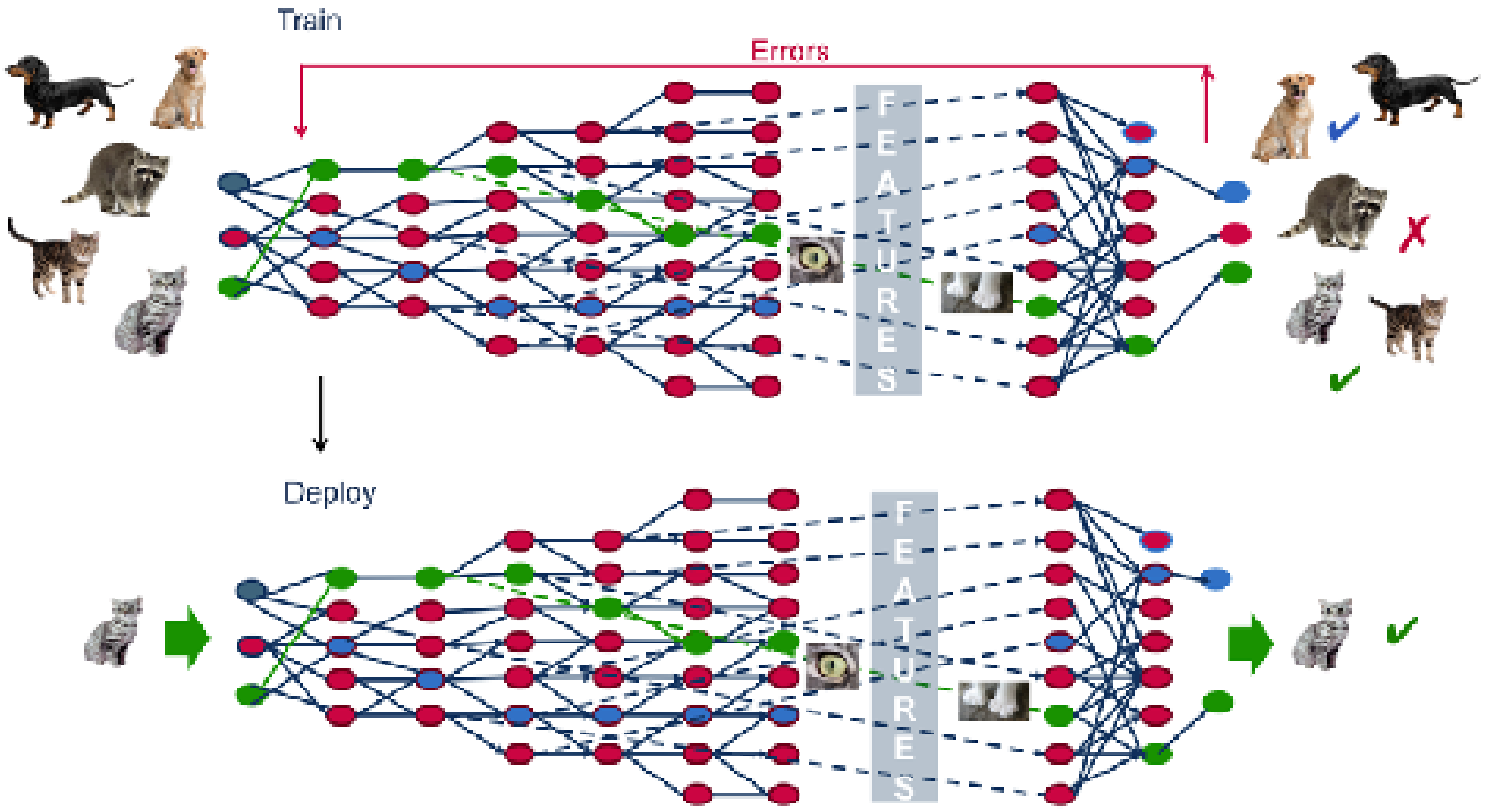
Phase

- Training: Model *learns* from data points and labels (e.g. animal image and animal class)
- Test: Given a new image (not a part of training) predicts the label (e.g. given an animal image, predicts its class)

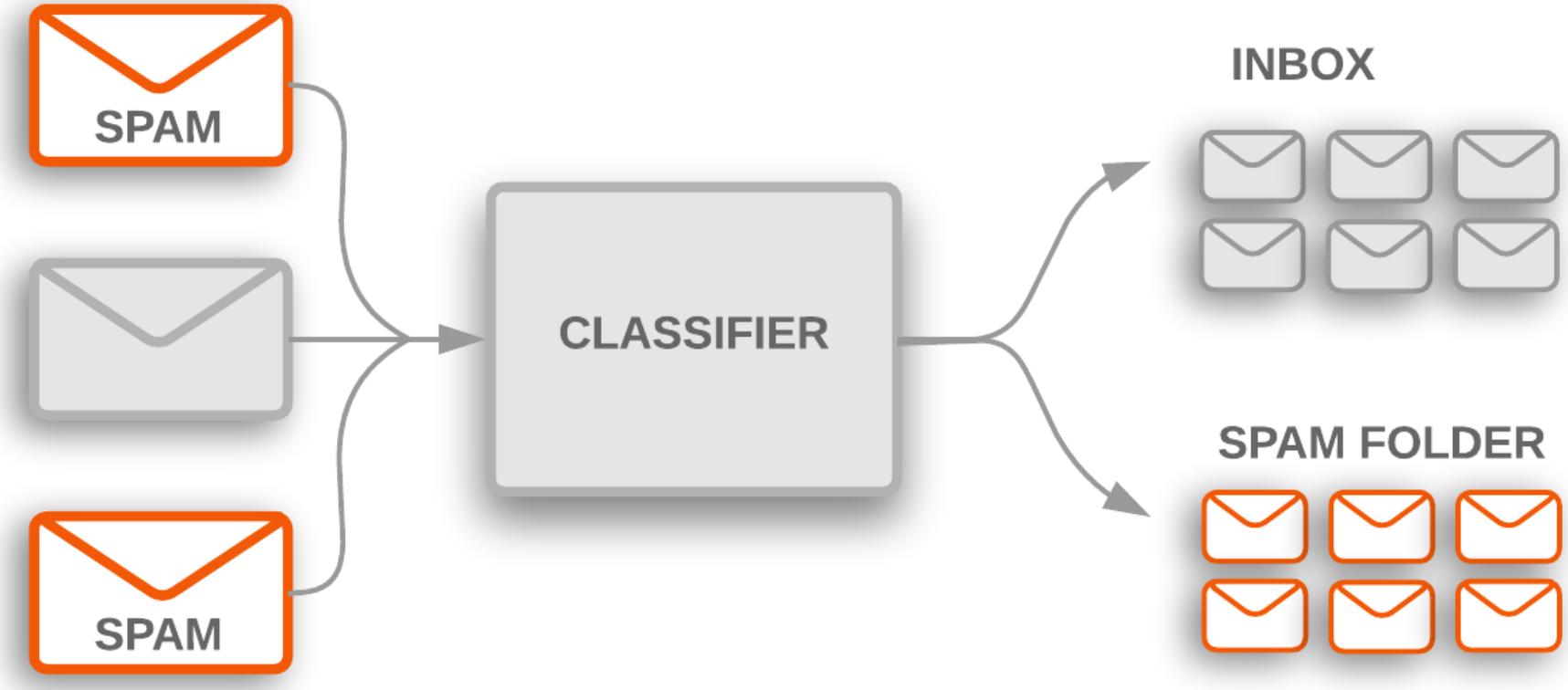
Classification Scheme



Classification



Classification: application



Classification: application

K-NEAREST NEIGHBOUR

K-Nearest Neighbour[2]

Training

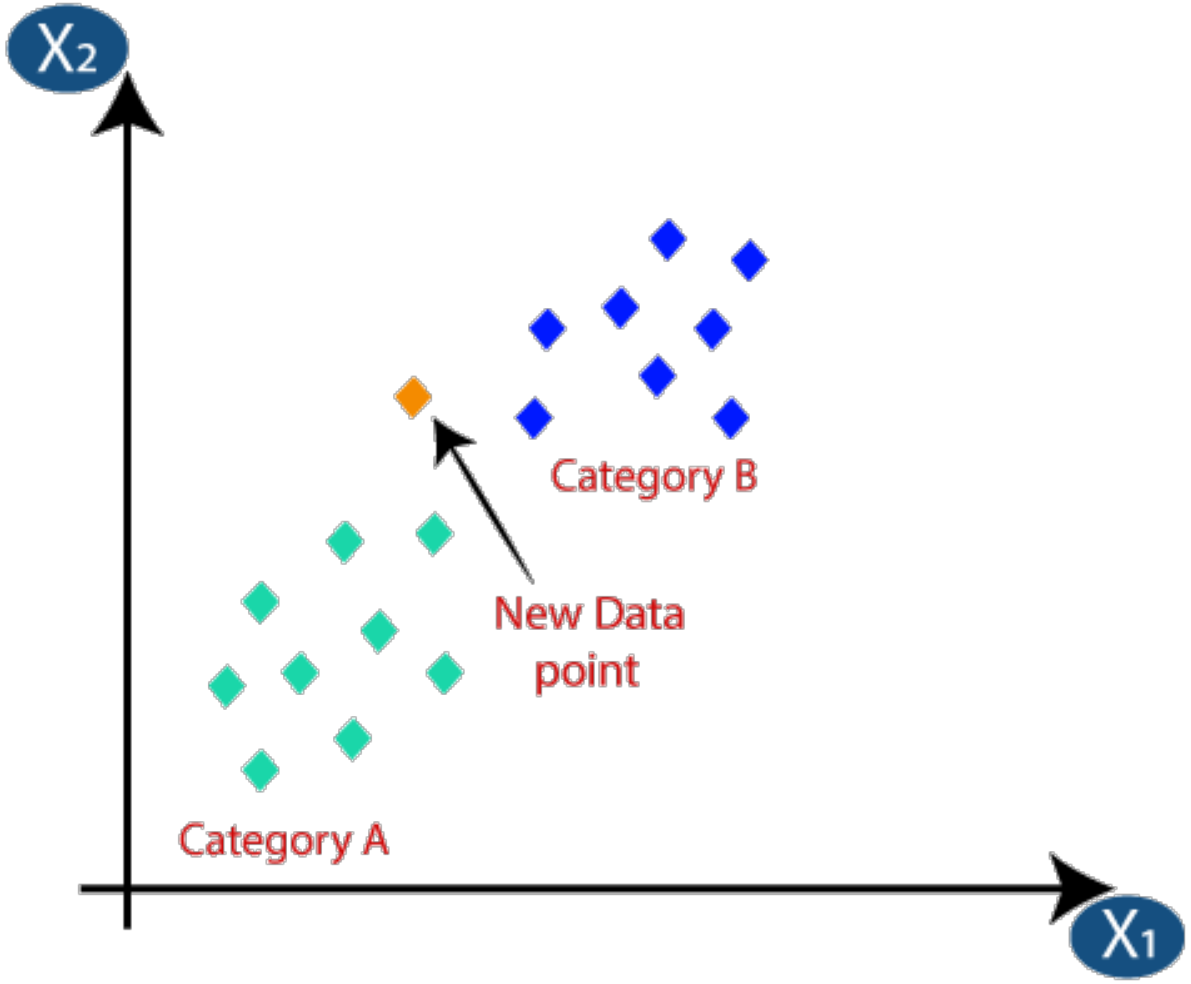
- Just stores the input data with labels
- It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset.

Test

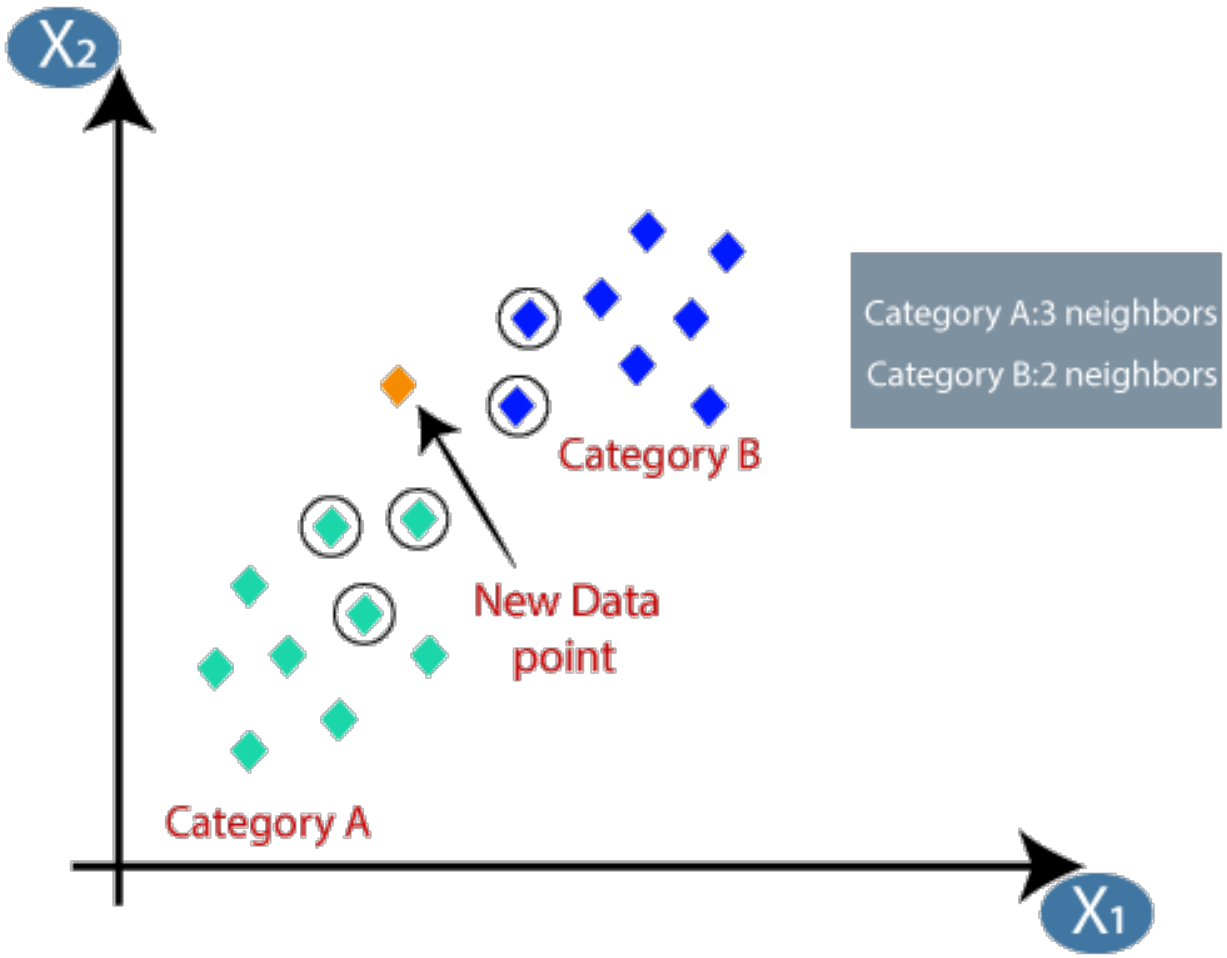
1. Step-1: Select the number K of the neighbours
2. Step-2: Calculate the distance of K number of neighbours with the new data point
3. Step-3: Take the K nearest neighbours as per the calculated distance.
4. Step-4: Among these K neighbors, count the number of the data points in each class/category.
5. Step-5: Assign the new data point to that class/category for which the number of the neighbor is maximum.

[2] https://www.javatpoint.com/k-nearest-neighbor-algorithm-for-machine-learning

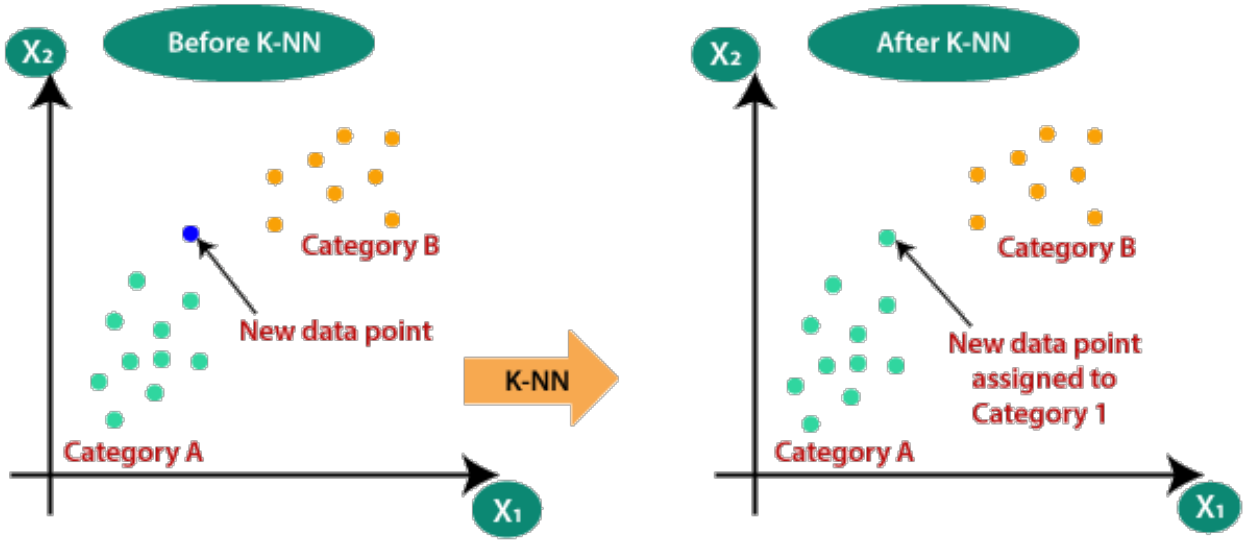
K-Nearest Neighbour



K-Nearest Neighbour



K-Nearest Neighbour



K-Nearest Neighbour

Advantage

Simple

Disadvantage

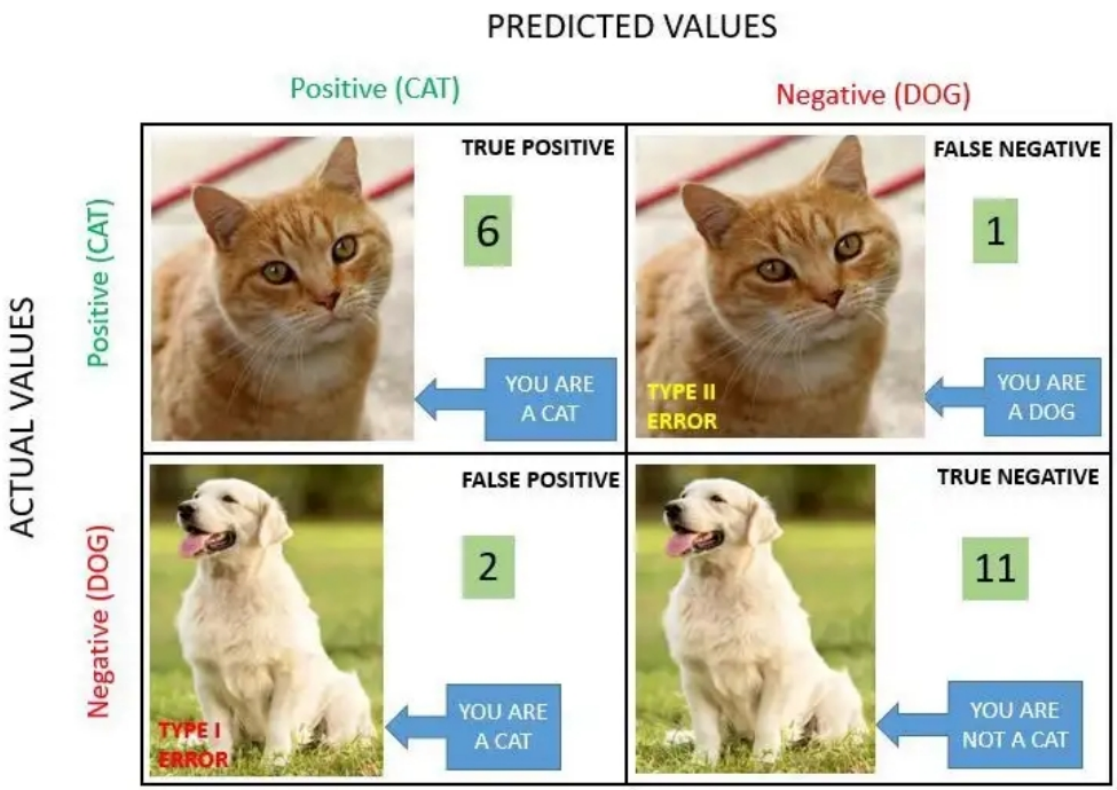
- Determining K can be a challenge
- Computationally expensive as involves the computation of distance of the new data point with all training points

Confusion Matrix[3]

		Predicted	
		Negative **(N)** -	Positive **(P)** +
Actual	Negative -	True Negative **(TN)**	**False Positive (FP) Type I Error**
	Positive +	**False Negative (FN) Type II Error**	True Positive **(TP)**

[3] https://medium.com/analytics-vidhya/what-is-a-confusion-matrix-d1c0f8feda5

Confusion Matrix[4]

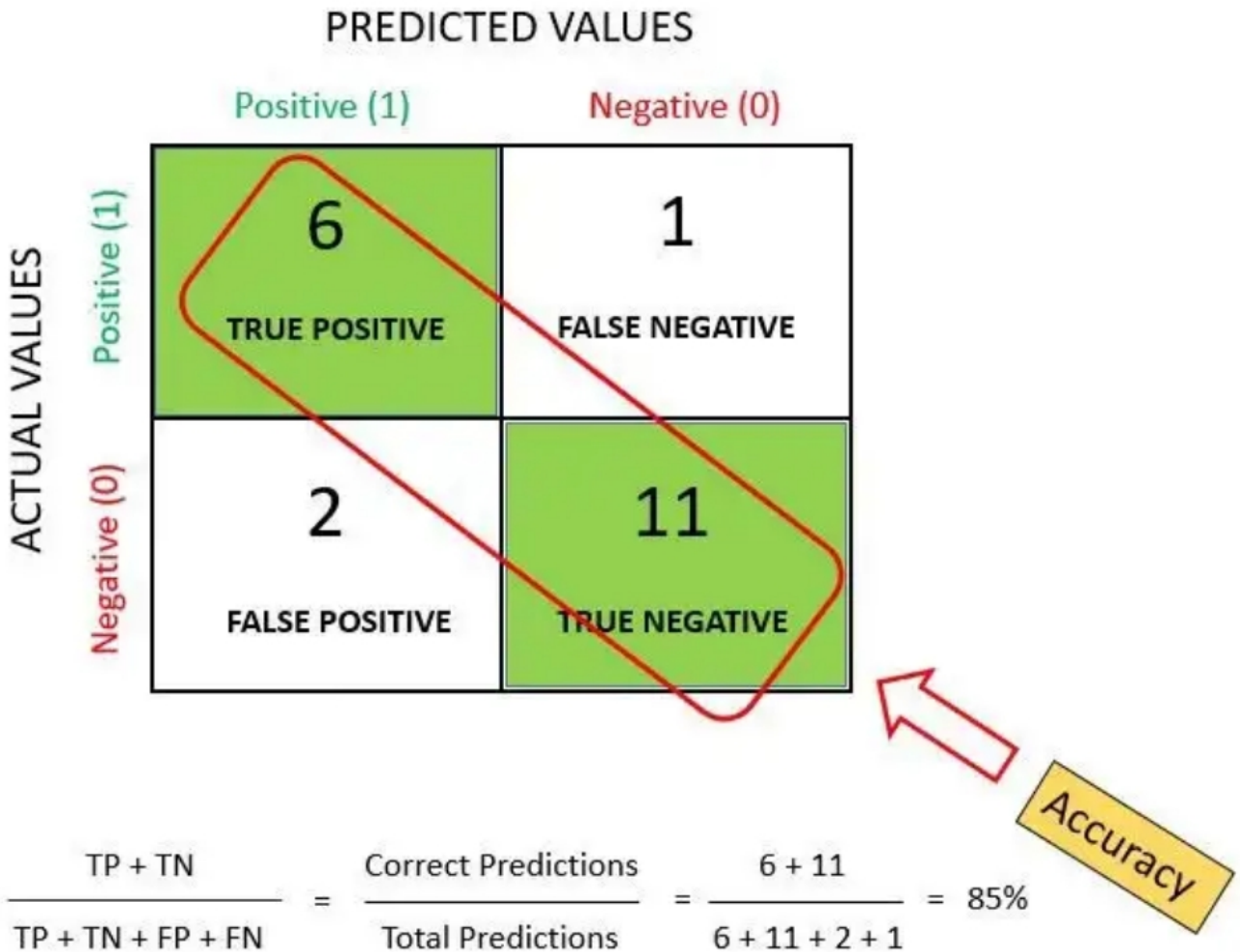


Classification Task

Positive: CAT, Negative: NOT CAT (DOG)

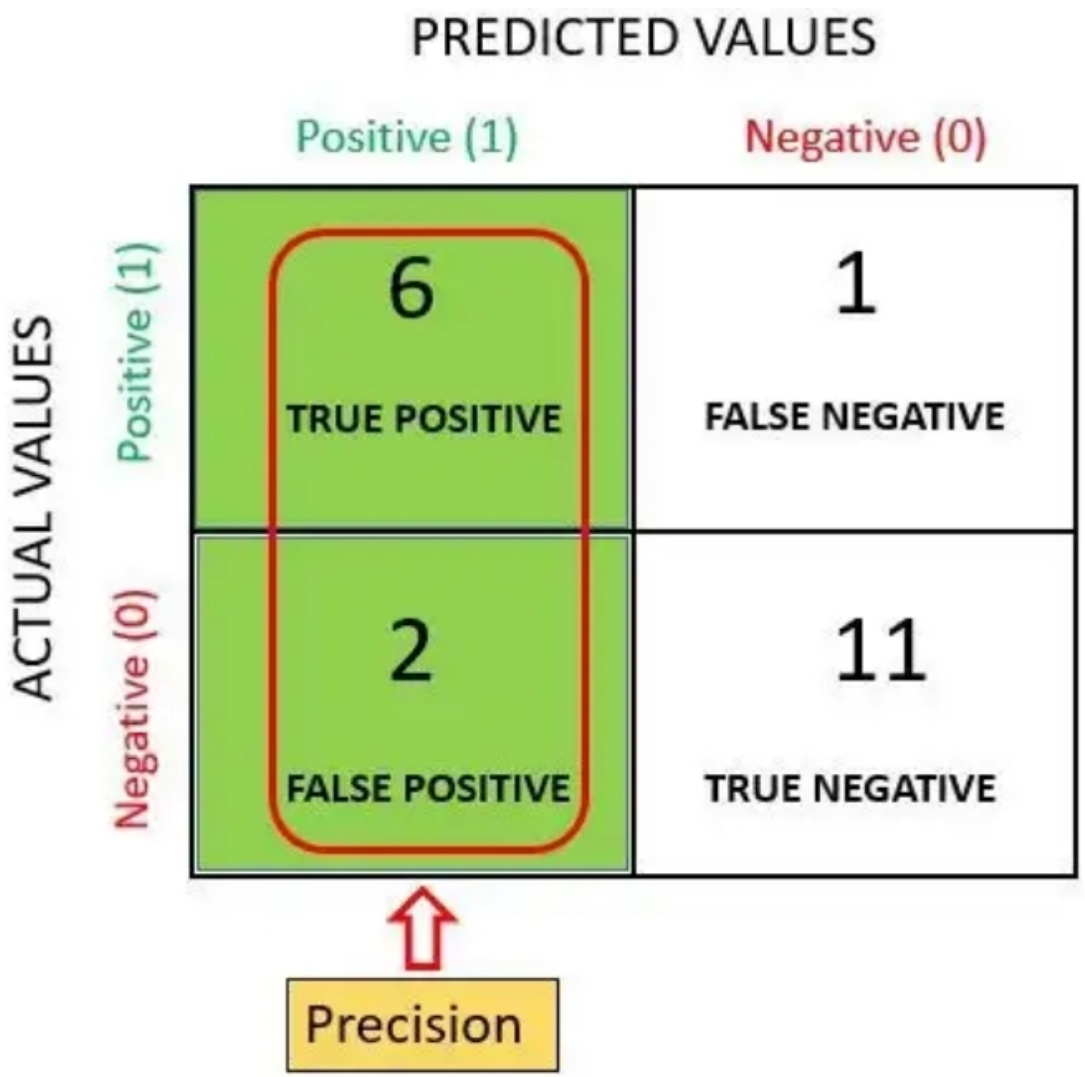
[4] https://medium.com/analytics-vidhya/what-is-a-confusion-matrix-d1c0f8feda5

Confusion Matrix: Accuracy



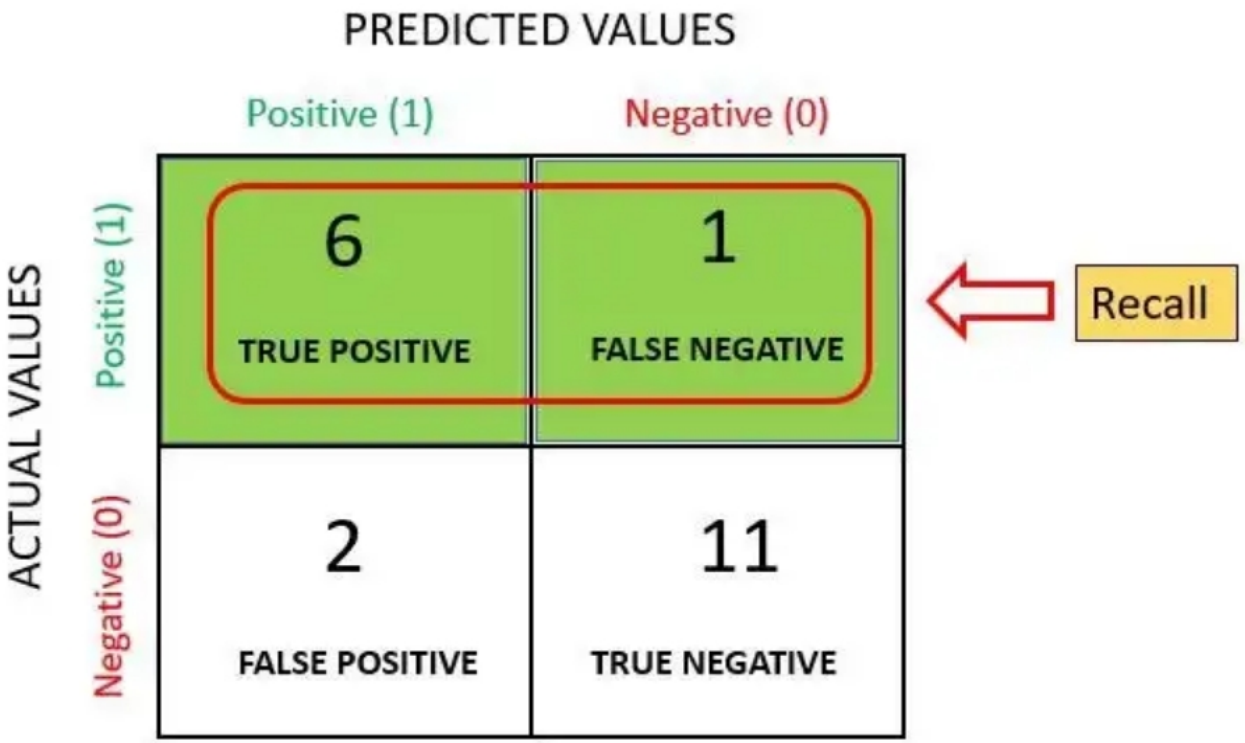
$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} = \frac{6 + 11}{6 + 11 + 2 + 1} = 85\%$$

Confusion Matrix: Precision



$$\text{Precision} = \frac{TP}{TP + FP} = \frac{\text{Predictions Actually Positive}}{\text{Total Predicted positive}} = \frac{6}{6+2} = 0.75$$

Confusion Matrix: Recall



$$\text{Recall} = \frac{TP}{TP + FN} = \frac{\text{Predictions Actually Positive}}{\text{Total Actual positive}} = \frac{6}{6+1} = 0.85$$

Confusion Matrix: F1-Score

$$\text{F1-Score} = 2 * \frac{(\text{Recall} * \text{Precision})}{(\text{Recall} + \text{Precision})} = 2 * \frac{(0.85 * 0.75)}{(0.85 + 0.75)} = 0.79$$

THANK YOU !!!